BOOST REGULATOR

REQUIRED														
-	-	Vin(Min)(V)	Vin(Max)(V)	Vout(V)	Iout(mA)	Iq(uA)	Output Type	Topology	Mounting Type	Available Supply	-	-	-	-
		1.8	3.2	3.45	50	100	Adj/Fixed	Boost	SMD	10000+				

PROPOSED

IC	Manufacturer	Vin(Min)(V)	Vin(Max)(V)	Vout(V)	Iout(mA)	Iq(uA)	Output Type	Topology	Mounting Type	Available Supply	Cost($)	Links	Datasheet	Remarks
TPS61023DRLR	Texas Instruments	0.5	5.5	2.2-5.5	1800	20	Adj	Boost	SMD	10000+	0.33	LCSC	TPS61023DRLR	This IC is preferred because of lowest cost and it has simple application circuit with less external components and high efficiency.
AP3015KTR-G1	Diodes Incorporated	1.2	12	1.23-34	350	1	Adj	Boost	SMD	10000+	0.76	Mouser	AP3015KTR-G1	It is designed for high power system with long range input with fewer external components. It is also cheap hence it is on Second Preference.
TPS613222ADBVT	Texas Instruments	0.9	5.5	1.8-5.5	900	6.5	Adj	Boost	SMD	10000+	1.16	Digikey	TPS613222ADBVR	This IC is of high efficiency with Synchronous configuration and fewer external components but its cost is greater than the above two, hence less preferred.
L6920DCTR	STMicroelectronics	0.6	5.5	3.3-5.5	800	11	Adj	Boost	SMD	10000+	2.88	Digikey	L6920DCTR	This IC is Synchronous with high efficiency and simple circuitry. It includes Over Current and Over Voltage Protection, but it is not preferred because it is expensive.
MAX17227AATA+T	Maxim Integrated	0.9	5.5	2.3-5.5	850	0.012	Adj	Boost	SMD	10000+	2.82	Mouser	MAX17227AATA+T	This IC is of high Efficiency with short circuit protection. It is equipped with simple application circuit, but is not preferred because of high cost.